Mathematics-I (M-I)
GTU # 3110014

Unit-1.2

Improper Integrals, Gamma Function & Beta Function

$$\int_{-\infty}^{\infty} f(x)\,dx$$

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OUTLINE

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- Method:7 ⟹ Improper Integral of Second Kind
- Method:8 ⟹ Convergence of Improper Integral of First Kind
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Improper Integral

- If the limit of integral is infinite(one or both) or/and integrand function is discontinuous for some value(s) on the interval of given integral then such integral is known as an improper integral.

Examples

(1). $\displaystyle\int_0^{\infty} \frac{1}{x^2+1}\,dx$ (2). $\displaystyle\int_{-\infty}^{0} x \sin x \, dx$ (3). $\displaystyle\int_{-\infty}^{\infty} e^x \, dx$

(4). $\displaystyle\int_0^{1} \frac{1}{x^2}\,dx$ (5). $\displaystyle\int_0^{\frac{\pi}{2}} \sec x \, dx$ (6). $\displaystyle\int_0^{3} \frac{1}{x-1}\,dx$

Types of Improper Integral

- There are three types of improper integrals

  (1). Improper integral of first kind.

  (2). Improper integral of second kind.

  (3). Improper integral of third kind(combination of 1^{st} & 2^{nd} kind).

Method:6 ➔ Improper Integral of First Kind

Improper Integral of First Kind

➡ For $\int_a^b f(x)\,dx$ either a or b or both(a & b) are infinite, then such integral is known as an improper integral of first kind.

Examples

(1). $\displaystyle\int_0^{\infty} \frac{dx}{x^2+1}$ (2). $\displaystyle\int_{-\infty}^{0} x \sin x \, dx$ (3). $\displaystyle\int_{-\infty}^{\infty} e^x \, dx$

Sub-types of Improper Integral of First Kind

(1). If $f(x)$ is continuous on $[a, \infty)$ then

$$\int_a^{\infty} f(x)\,dx = \lim_{b \to \infty} \int_a^{b} f(x)\,dx.$$

(2). If $f(x)$ is continuous on $(-\infty, b]$ then

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a \to -\infty} \int_a^{b} f(x)\,dx.$$

(3). If $f(x)$ is continuous on $(-\infty, \infty)$ then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{t} f(x)\,dx + \int_{t}^{\infty} f(x)\,dx = \lim_{a \to -\infty} \int_a^{t} f(x)\,dx + \lim_{b \to \infty} \int_t^{b} f(x)\,dx.$$

Improper Integral of First Kind

Example 1 Evaluate $\int_0^\infty \frac{1}{x^2+1}\,dx$.

Solution

$$\int_0^\infty \frac{1}{x^2+1}\,dx = \lim_{b\to\infty}\int_0^b \frac{1}{x^2+1}\,dx = \lim_{b\to\infty}\left[\tan^{-1}(x)\right]_0^b$$

$$= \lim_{b\to\infty}\left[\tan^{-1}(b) - \tan^{-1}(0)\right]$$

$$= \lim_{b\to\infty}\left[\tan^{-1}(b)\right] - \tan^{-1}(0) = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

Hence, $\boxed{\int_0^\infty \frac{1}{x^2+1}\,dx = \frac{\pi}{2}}$.

Improper Integral of First Kind

Example 5 Evaluate $\int_{-\infty}^{0} e^{2x}\, dx$.

Solution

$$\int_{-\infty}^{0} e^{2x}\, dx = \lim_{a \to -\infty} \int_{a}^{0} e^{2x}\, dx = \lim_{a \to -\infty} \left[\frac{e^{2x}}{2}\right]_a^0$$

$$= \lim_{a \to -\infty} \left[\frac{e^{0}}{2} - \frac{e^{2a}}{2}\right] = \frac{1}{2} - \lim_{a \to -\infty} \left[\frac{e^{2a}}{2}\right]$$

$$= \frac{1}{2} - 0 = \frac{1}{2}$$

Hence, $\boxed{\int_{-\infty}^{0} e^{2x}\, dx = \frac{1}{2}}$.

Improper Integral of First Kind

Example 9 Evaluate $\int_{-\infty}^{\infty} e^x \, dx$.

Solution

$$\int_{-\infty}^{\infty} e^x \, dx = \int_{-\infty}^{t} e^x \, dx + \int_{t}^{\infty} e^x \, dx$$

$$= \lim_{a \to -\infty} \int_{a}^{t} e^x \, dx + \lim_{b \to \infty} \int_{t}^{b} e^x \, dx$$

$$= \lim_{a \to -\infty} [e^x]_a^t + \lim_{b \to \infty} [e^x]_t^b$$

Improper Integral of First Kind

Solution Continue

$$\lim_{a \to -\infty} [e^x]_a^t + \lim_{b \to \infty} [e^x]_t^b$$

$$= \cancel{e^t} - \lim_{a \to -\infty} e^a + \lim_{b \to \infty} e^b - \cancel{e^t}$$

$$= 0 + \infty$$

$$= \infty$$

Hence, $\boxed{\int_{-\infty}^{\infty} e^x \, dx = \infty}$.

Method:7 ➔ Improper Integral of Second Kind

Improper Integral of Second Kind

➡ For $\int_a^b f(x)\,dx$, $f(x)$ become discontinuous at $x = a$ or $x = b$ or at finite number of points in the interval (a, b), then such an integral is known as improper integral of second kind.

Examples

(1). $\int_0^1 \frac{1}{x^2}\,dx$ (2). $\int_0^{\frac{\pi}{2}} \sec x\,dx$ (3). $\int_0^3 \frac{1}{x-1}\,dx$

Sub-types of Improper Integral of Second Kind

(1). If $x = a$ is the point of discontinuity for $f(x)$ then

$$\int_a^b f(x)\,dx = \lim_{t \to a} \int_t^b f(x)\,dx.$$

(2). If $x = b$ is the point of discontinuity for $f(x)$ then

$$\int_a^b f(x)\,dx = \lim_{t \to b} \int_a^t f(x)\,dx.$$

(3). If $x = c \in (a, b)$ is the point of discontinuity for $f(x)$ then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \lim_{t \to c} \int_a^t f(x)\,dx + \lim_{t \to c} \int_t^b f(x)\,dx.$$

Improper Integral of Second Kind

Example 9 Evaluate $\displaystyle\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx$.

Solution

$$\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx = \int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx + \int_1^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx$$

$$= \lim_{t\to 1} \int_0^t (x-1)^{-\frac{2}{3}}\,dx + \lim_{t\to 1} \int_t^3 (x-1)^{-\frac{2}{3}}\,dx$$

$$= \lim_{t\to 1} \left[\frac{(x-1)^{\frac{1}{3}}}{\frac{1}{3}}\right]_0^t + \lim_{t\to 1} \left[\frac{(x-1)^{\frac{1}{3}}}{\frac{1}{3}}\right]_t^3$$

Improper Integral of Second Kind

Solution Continue

$$\lim_{t\to 1}\left[3(x-1)^{\frac{1}{3}}\right]_0^t + \lim_{t\to 1}\left[3(x-1)^{\frac{1}{3}}\right]_t^3$$

$$= \lim_{t\to 1}\left[3(t-1)^{\frac{1}{3}}\right] - 3(-1)^{\frac{1}{3}} + 3(2)^{\frac{1}{3}} - \lim_{t\to 1}\left[3(t-1)^{\frac{1}{3}}\right]$$

$$= 0 - 3(-1)^{\frac{1}{3}} + 3(2)^{\frac{1}{3}} - 0$$

$$= 3\left[(2)^{\frac{1}{3}} - (-1)^{\frac{1}{3}}\right]$$

Hence, $\boxed{\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx = 3\left[(2)^{\frac{1}{3}} - (-1)^{\frac{1}{3}}\right]}$.

Convergence or Divergence of Improper Integral

- If integral has finite value then we can say that given integral is convergent.
- If integral has infinite value then we can say that given integral is divergent.

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Method:8 → Convergence of Improper Integral of First Kind

Convergence of Improper Integral of First Kind

Example 8 Investigate the convergence of $\int_{5}^{\infty} \frac{5x}{(1+x^2)^3}\,dx$.

Solution

$$\int_{5}^{\infty} \frac{5x}{(1+x^2)^3}\,dx = \frac{5}{2}\int_{5}^{\infty} (2x)(1+x^2)^{-3}dx$$

$$= \frac{5}{2}\lim_{b\to\infty}\int_{5}^{b} (1+x^2)^{-3}(2x)\,dx = \frac{5}{2}\lim_{b\to\infty}\left[\frac{(1+x^2)^{-2}}{-2}\right]_5^b$$

$$\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + c$$

Convergence of Improper Integral of First Kind

Solution Continue

$$-\frac{5}{4}\lim_{b\to\infty}\left[\frac{1}{(1+x^2)^2}\right]_5^b$$

$$=-\frac{5}{4}\left[\lim_{b\to\infty}\frac{1}{(1+b^2)^2}\right]+\frac{5}{4}\left[\frac{1}{(1+(5)^2)^2}\right]$$

$$=0+\frac{5}{4}\left[\frac{1}{(26)^2}\right]=\frac{5}{2704}$$ Which is finite.

Hence, $\boxed{\int_5^\infty \frac{5x}{(1+x^2)^3}\,dx \text{ is convergent.}}$

P-integral Test

- $\int_a^{\infty} \frac{1}{x^p}\,dx$ is convergent if $p > 1$ & divergent if $p \leq 1$.
- $\int_0^{a} \frac{1}{x^p}\,dx$ is convergent if $p < 1$ & divergent if $p \geq 1$.

Direct Comparison Test

- If $f(x)$ and $g(x)$ are continuous on $[a, \infty)$ and $0 \leq f(x) \leq g(x)$ for all $x \geq a$ then

(1). If $\int_a^\infty g(x)\,dx$ is convergent then $\int_a^\infty f(x)\,dx$ is convergent.

(2). If $\int_a^\infty f(x)\,dx$ is divergent then $\int_a^\infty g(x)\,dx$ is divergent.

Convergence of Improper Integral of First Kind

Example 5(1) Check the convergence of $\int_{1}^{\infty} \frac{\sin^2 x}{x^2}\, dx$.

Solution We know that $-1 \leq \sin x \leq 1$

$$\Rightarrow 0 \leq \sin^2 x \leq 1$$

$$\Rightarrow 0 \leq \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2}, \forall x \in [1, \infty)$$

We know that $\int_{1}^{\infty} \frac{1}{x^2}\, dx$ is convergent by p-integral test.

P-integral Test

$\int_{a}^{\infty} \frac{1}{x^p}\, dx$ is convergent if $p > 1$ & divergent if $p \leq 1$.

Solution Continue

Direct Comparison Test

Hence, by direct comparison test $\int_{1}^{\infty} \frac{\sin^2 x}{x^2}\, dx$ is convergent.

➡ If $f(x)$ and $g(x)$ are continuous on $[a, \infty)$ and $0 \le f(x) \le g(x)$ for all $x \ge a$ then

(1). If $\int_{a}^{\infty} g(x)\, dx$ is convergent then $\int_{a}^{\infty} f(x)\, dx$ is convergent.

Limit Comparison Test

- If $f(x)$ and $g(x)$ are positive and continuous on $[a, \infty)$ and $L = \lim_{x \to \infty} \frac{f(x)}{g(x)}$ then

(1). If $0 < L < \infty$ then $\int_a^\infty f(x)\,dx$ and $\int_a^\infty g(x)\,dx$ both are convergent or divergent together.

(2). If $L = 0$ and $\int_a^\infty g(x)\,dx$ is convergent then $\int_a^\infty f(x)\,dx$ is convergent.

(3). If $L = \infty$ and $\int_a^\infty g(x)\,dx$ is divergent then $\int_a^\infty f(x)\,dx$ is divergent.

Example 10 Check the convergence of $\int_{4}^{\infty} \frac{3x+5}{x^4+7}\, dx$.

Solution

Let $\int_{4}^{\infty} \frac{3x+5}{x^4+7}\, dx = \int_{4}^{\infty} f(x)\, dx$

Consider $g(x) = \frac{x}{x^4} = \frac{1}{x^3}$

Now, $L = \lim_{x \to \infty} \frac{f(x)}{g(x)} = \lim_{x \to \infty} \frac{\frac{3x+5}{x^4+7}}{\frac{1}{x^3}} = \lim_{x \to \infty} \frac{(x^3)(3x+5)}{x^4+7}$

Convergence of Improper Integral of First Kind

Solution Continue

$$\lim_{x \to \infty} \frac{(x^3)(3x+5)}{x^4+7}$$

$$= \lim_{x \to \infty} \frac{\cancel{x^4}\left(3+\frac{5}{x}\right)}{\cancel{x^4}\left(1+\frac{7}{x^4}\right)} = \lim_{x \to \infty} \frac{\left(3+\frac{5}{x}\right)}{\left(1+\frac{7}{x^4}\right)} = 3$$

$$\Rightarrow 0 < L = 3 < \infty$$

We know that $\int_4^{\infty} \frac{1}{x^3}\, dx$ is convergent by p-integral test.

Convergence of Improper Integral of First Kind

Solution Continue

Hence, by limit comparison test $\boxed{\int_4^{\infty} \frac{3x+5}{x^4+7}\,dx \text{ is convergent.}}$

Method:9 → Convergence of Improper Integral of Second Kind

Example 5 Define improper integral of both the kinds and check the convergence of

$$\int_0^3 \frac{dx}{\sqrt{9-x^2}}.$$

Solution

- For $\int_a^b f(x)\,dx$ either a or b or both(a & b) are infinite, then such integral is known as an improper integral of first kind.

Convergence of Improper Integral of Second Kind

Solution Continue

- For $\int_a^b f(x)\,dx$, $f(x)$ become discontinuous at $x = a$ or $x = b$ or at finite number of points in the interval (a, b), then such an integral is known as improper integral of second kind.

$$\int_0^3 \frac{dx}{\sqrt{9 - x^2}} = \lim_{b \to 3} \int_0^b \frac{dx}{\sqrt{9 - x^2}} = \lim_{b \to 3} \left[\sin^{-1}\left(\frac{x}{3}\right)\right]_0^b$$

Convergence of Improper Integral of Second Kind

Solution Continue

$$\lim_{b \to 3}\left[\sin^{-1}\left(\frac{x}{3}\right)\right]_0^b$$

$$= \lim_{b \to 3}\sin^{-1}\left(\frac{b}{3}\right) - \sin^{-1}(0)$$

$$= \sin^{-1}(1) - \sin^{-1}(0) = \frac{\pi}{2} - 0$$

$$= \frac{\pi}{2}$$ Which is finite.

Hence, $\boxed{\int_0^3 \frac{dx}{\sqrt{9-x^2}} \text{ is convergent.}}$

Gamma Function

- If $n > 0$, then Gamma function is defined by the integral $\int_0^\infty e^{-x} x^{n-1}\, dx$

  and it is denoted by $\Gamma(n)$. i.e. $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\, dx$.

Properties of Gamma Function

(1). Reduction formula for Gamma function is $\Gamma(n+1) = \Gamma(n)$; where $n > 0$.

(2). If n is positive integer, then $\Gamma(n+1) = n!$.

(3). Second form of Gamma function is $\int_0^\infty e^{-x^2} x^{2m-1}\, dx = \frac{1}{2}\Gamma(n)$.

(4). $\Gamma\left(n+\frac{1}{2}\right) = \frac{(2n)!\sqrt{\pi}}{n!\,4^n}$ for $n = 0, 1, 2, 3, \dots$. For $n = 0, \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

For $n = 1, \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$ For $n = 2, \Gamma\left(\frac{5}{2}\right) = \frac{3\sqrt{\pi}}{4}$

Beta Function

- If $m > 0, n > 0,$ then **Beta** function is defined by the integral

$\int_0^1 x^{m-1}(1-x)^{n-1}\,dx$ and it is denoted by $\beta(m, n)$ or $B(m, n)$.

i.e. $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$.

Properties of Beta Function

(1). Beta function is a symmetric function. i.e. $\beta(m, n) = \beta(n, m)$, where $m, n > 0.$

(2). $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta.$

(3). $\int_0^{\frac{\pi}{2}} \sin^p\theta \cos^q\theta \, d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right).$

(4). $\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} \, dx.$

Properties of Beta Function

(5). Relation between Beta and Gamma function, $\beta(m, n) = \dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

Example 2 Define Gamma function and evaluate $\int_0^\infty e^{-x^2}\,dx$.

Solution

- If $n > 0$, then **Gamma** function is defined by the integral $\int_0^\infty e^{-x}x^{n-1}\,dx$

  and it is denoted by $\Gamma(n)$. i.e. $\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}\,dx$.

Example on Gamma Function and Beta Function

Solution Continue $\int_0^{\infty} e^{-x^2}dx$

- Consider $x^2 = t \Rightarrow x = \sqrt{t}$

$$\Rightarrow 2x\,dx = dt \Rightarrow dx = \frac{dt}{2x} = \frac{dt}{2\sqrt{t}}$$

- Now $x \to 0 \Rightarrow t \to 0$ & $x \to \infty \Rightarrow t \to \infty$

$$\Rightarrow \int_0^{\infty} e^{-x^2}dx = \int_0^{\infty} \frac{e^{-t}dt}{2\sqrt{t}} = \frac{1}{2}\int_0^{\infty} e^{-t}\,t^{-\frac{1}{2}}dt$$

Example on Gamma Function and Beta Function

Solution Continue

$$\frac{1}{2}\int_0^\infty e^{-t}\, t^{\frac{1}{2}-1} dt$$

$$= \frac{1}{2}\Gamma\left(\frac{1}{2}\right) \quad \left(\because \Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\, dx\right)$$

$$= \frac{\sqrt{\pi}}{2}$$

Hence, $\boxed{\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}}$

Example on Gamma Function and Beta Function

Example 8 Find $\beta\left(\frac{7}{2},\frac{5}{2}\right)$.

Solution We know that $\beta(m,n)=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

$$\Rightarrow \beta\left(\frac{7}{2},\frac{5}{2}\right)=\frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{7}{2}+\frac{5}{2}\right)}=\frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{5}{2}\right)}{\Gamma(6)}$$

$$=\frac{\left(\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\sqrt{\pi}\right)\left(\frac{3}{2}\cdot\frac{1}{2}\cdot\sqrt{\pi}\right)}{(5)!}$$

Example on Gamma Function and Beta Function

Solution Continue

$$= \frac{45\pi}{32(120)}$$

$$= \frac{3\pi}{256}$$

Hence, $\boxed{\beta\left(\frac{7}{2}, \frac{5}{2}\right) = \frac{3\pi}{256}}.$

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THANK YOU

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